

Measurement Invariance Power App

Shiny app and R functions for computing analytic power of chi-square difference tests of **scale-level** (parameter-total) vs **item-level** measurement invariance in multiple-group one-factor CFA, generalizing the code behind the manuscript in Old Jordan Files for Claude/.

Run the app

```
shiny::runApp("path/to/ShinyApp")
```

Requires: lavaan, shiny, bslib, ggplot2, DT.

Files

File	Contents
app.R	Shiny UI/server only (no statistics live here)
R/models.R	Population specification (default_params, apply_noninvariance, build_structured_params, structured_sweep, pop_moments) and lavaan model generation for any number of items p and groups G (make_item_model, make_scale_model)
R/power.R	mi_power() / mi_power_sweep(): MacCallum-Browne-Cai (2006) analytic power for metric/scalar/strict tests, both approaches, returned as tidy data frames
R/plots.R	plot_power(): ggplot2 power curves (solid = scale-level, dashed = item-level; color = overlaid condition)
validation/validate_manuscript.R	Reproduces every population RMSEA annotated in manuscript Figures 1-6 and 8 and checks power-formula equivalence with the original pipeline. Run: Rscript validation/validate_manuscript.R (all 44 checks pass)

Scripting the functions directly

```
invisible(lapply(list.files("R", full.names = TRUE), source))
library(lavaan)

# Manuscript Fig 3 (all panels in one plot):  $k = 7..2$  non-invariant intercepts
conds <- structured_sweep(
  list(p = 8, n_groups = 2, type = "intercepts", n_ni = 3, magnitude = .1),
  sweep_var = "n_ni", values = 7:2)
res <- mi_power_sweep(conds, tests = c("metric", "scalar"))
plot_power(res$curves)
res$details # population RMSEA, test df, N needed for 80% power
```

Reproducing the manuscript figures in the app

Figure numbers refer to “figures with captions.pdf” (the output of `figure markdown4.Rmd`). Each 6-panel manuscript figure becomes **one** overlay plot in the app: the swept conditions (colors) correspond to the panels (a)-(f). Read the facet named in the “Read” column; the other facet shows the same population under the other test.

Common settings for every figure (the app defaults): *Model*: 2 groups, $p = 8$ items, base loading .7, base residual variance .51. *Test settings*: N per group 50-1000, $\alpha = .05$. Tick “Also test strict invariance” for the residual-variance figures (it turns on automatically when the non-invariance type is set to residual variances).

The manuscript places non-invariance on different item positions than the app (which always uses the last k items); because all other items are exchangeable this yields identical fit statistics and power.

Fig	Non-invariance panel	Sweep panel	Read
1 (a,b)	Type: Loadings; count: 6; magnitude: per-item .1; direction: Alternating +/-	Vary: Magnitude; values: 0.1, 0.2	Metric facet: item-level (dashed) has power, scale-level (solid) stays at .05
1 (c,d)	as above but Type: Intercepts	same	Scalar facet
1 (e,f)	as above but Type: Residual variances	same	Strict facet
2	Type: Loadings; magnitude: per-item .1; direction: All positive	Vary: Number of non-invariant items; range 2-7	Metric facet (panels a-f = 7,6,5,4,3,2 non-invariant)
3	Type: Intercepts; otherwise as Fig 2	same	Scalar facet
4	Type: Residual variances; otherwise as Fig 2	same	Strict facet
5	Type: Loadings; magnitude: Total across items = .6; direction: All positive	Vary: Number of non-invariant items; range 1-6	Metric facet (per-item delta becomes .6/k: .1, .12, .15, .2, .3, .6)
6	Type: Intercepts; otherwise as Fig 5	same	Scalar facet
7	Type: Residual variances; otherwise as Fig 5	same	Strict facet
8	Type: Loadings; count: 3; magnitude: per-item .2; direction: All positive	Vary: Scale length (p); values: 4, 6, 8, 10, 12, 14	Metric facet
9	Type: Intercepts; otherwise as Fig 8	same	Scalar facet
10	Type: Residual variances; otherwise as Fig 8	same	Strict facet

Notes: - Fig 1’s panels differ only in magnitude (.1 vs .2), so the app shows the two panels of each pair as two colored curves in one facet. With the alternating direction and an even count the parameter totals are equal across groups (a non-impactful violation), which is the point of Study 1. - The manuscript

curves show power of the test matching the manipulated parameter type only; the app additionally plots the other tests for the same population. For Fig 2/5/8 (loadings) the scalar facet is flat at alpha because intercepts remain invariant, and vice versa. - Exact expected rmsea values for Figs 1-6 and 8 are asserted in `validation/validate_manuscript.R` (the Details tab's rmsea column should match the figure annotations to 3 decimals).

Method notes / differences from the original scripts

- Noncentrality is computed from the population minimum discrepancy: `nonc = (N_total - G) * (F0_constrained - F0_baseline)`, which for $G = 2$ is numerically identical to the manuscript's `n * df * rmsea^2` but is correct for any G and independent of lavaan's multi-group RMSEA convention. Baselines are one invariance level lower (configural -> metric -> scalar -> strict), fit with the same approach.
- Test df are taken from the fitted models. This silently fixes a bug in the original `power.plot.doer.weak/.strict`, where the critical value used `nrow(pop.cov1-1) = p` df instead of $p - 1$ for the metric test (a small effect, e.g. power .797 vs .819 at $n = 300$ in Fig 1a).
- Scale-level models for G groups impose $G - 1$ total-equality constraints per parameter class (plus the first-intercept anchor for scalar).
- Population latent variance is 1 and latent mean 0 in all groups; ad-hoc effect sizes (`beta_spurious`, `d_spurious`) are intentionally omitted.